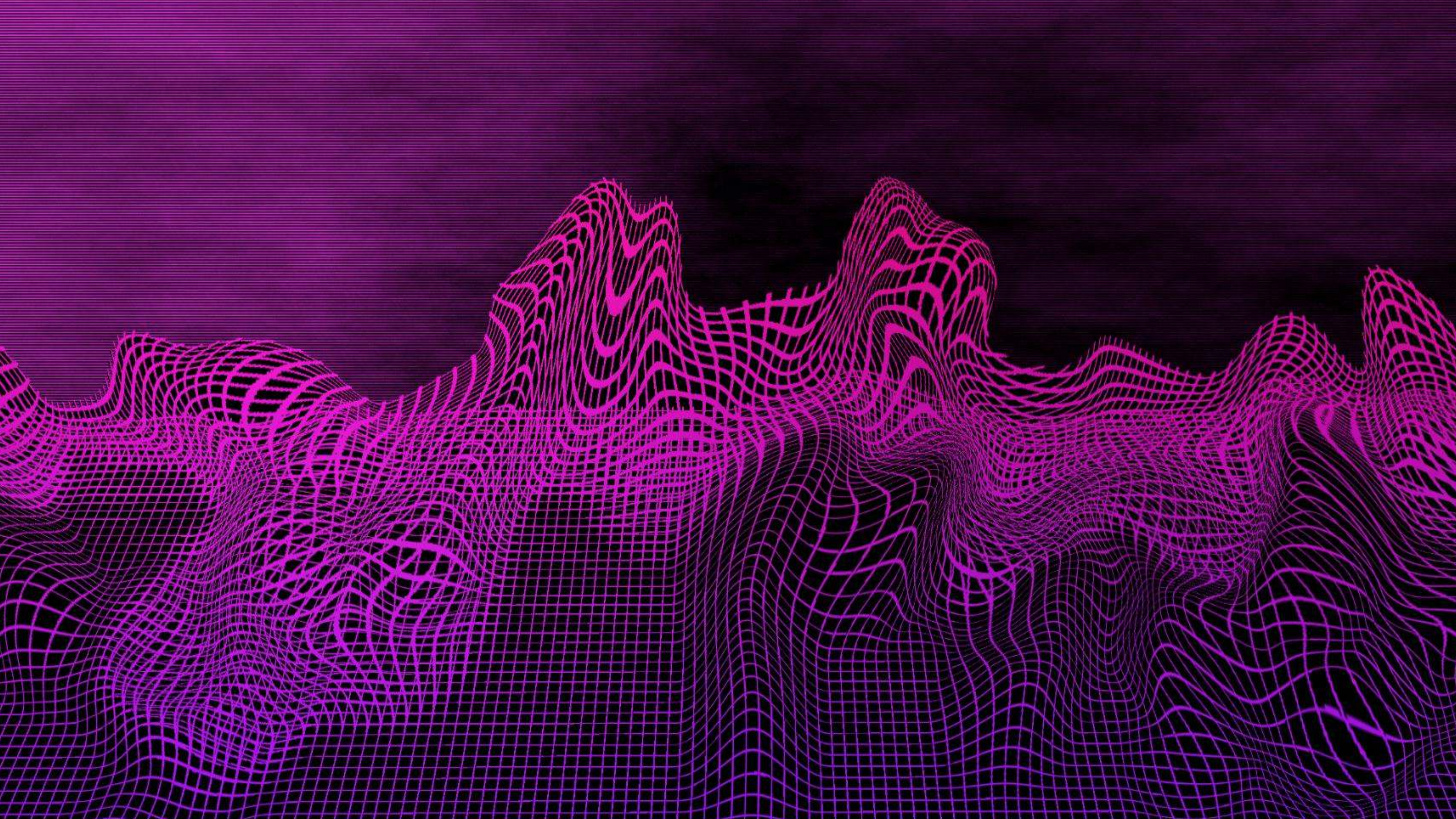


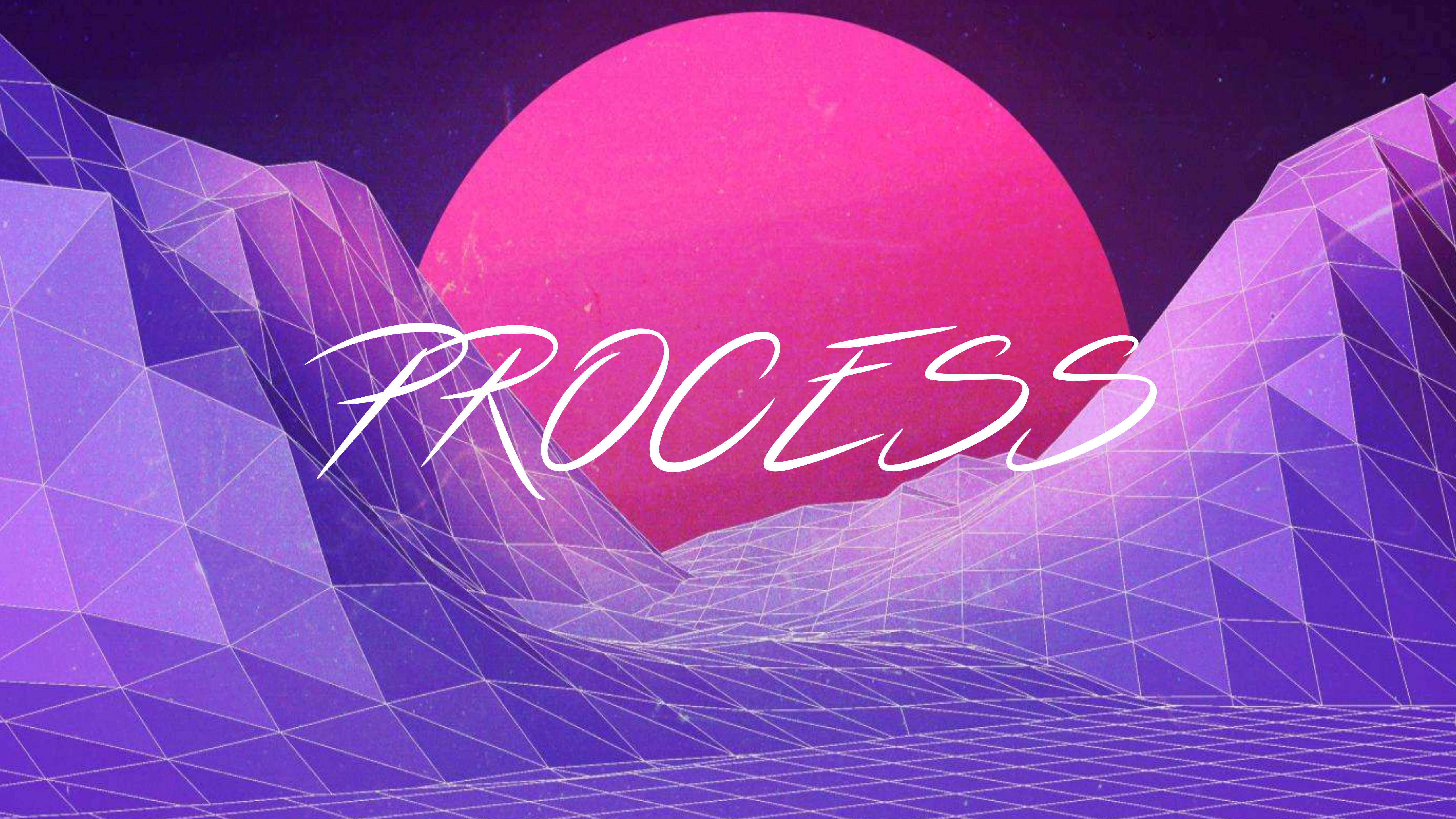


THAT ONE,
FOR NOW

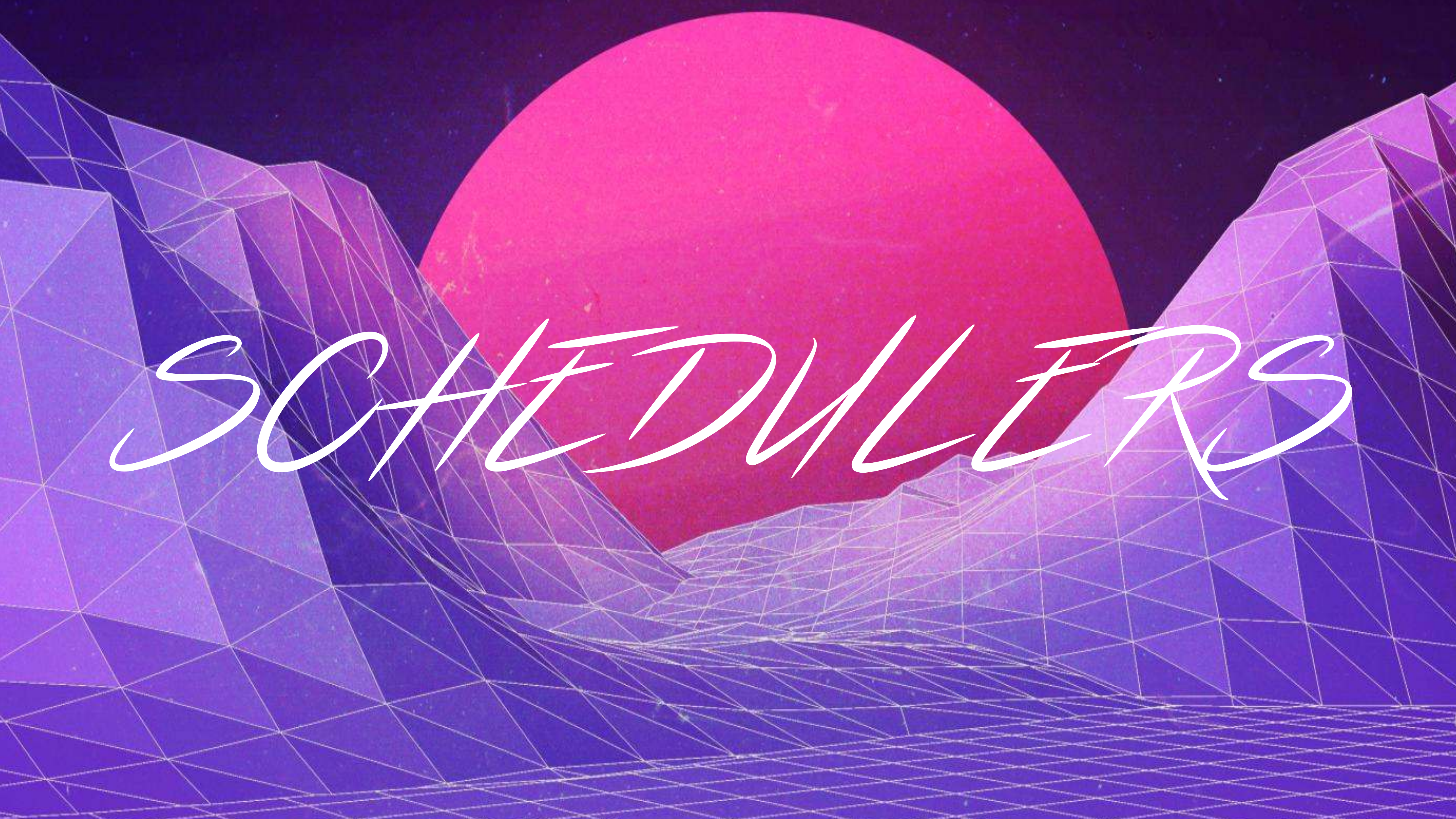
@bgmarx



*RHODENIX
REQUEST*



PROCESS



SCHEDULERS

The background features a dark purple field with a large, solid pink circle in the upper center. On either side of the circle are jagged, low-poly mountain-like shapes in shades of purple. The foreground is filled with a dense, low-poly grid of lines that recede into the distance, creating a sense of depth.

SINGLE SCHEDULER

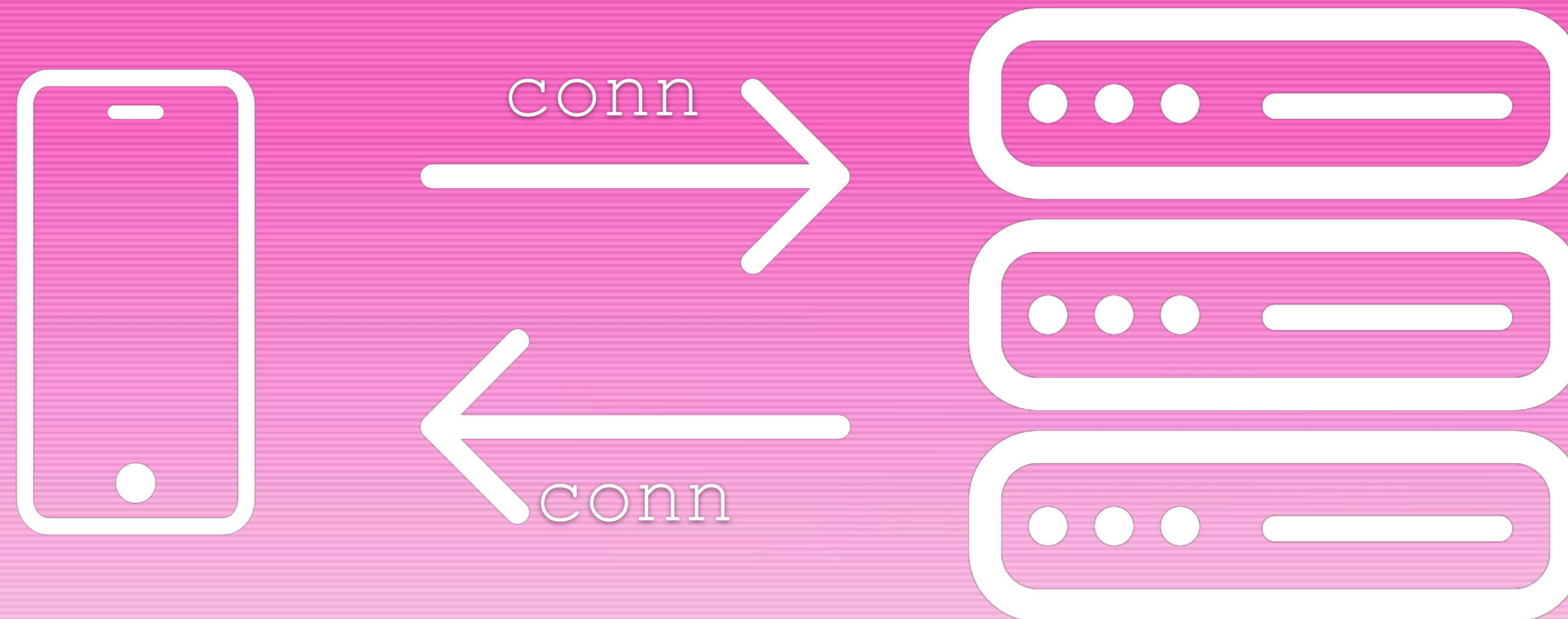
The background features a dark purple field with a large, solid pink circle in the upper center. On either side of the circle are complex, low-poly geometric structures in shades of purple and blue, resembling crystalline or architectural forms. The entire scene is overlaid with a fine, white grid of lines that create a sense of depth and perspective.

MULTIPLE
SCHEDULES
(SNP)

*RHODENIX
REQUEST*



transformations



endpoint

plugs

router

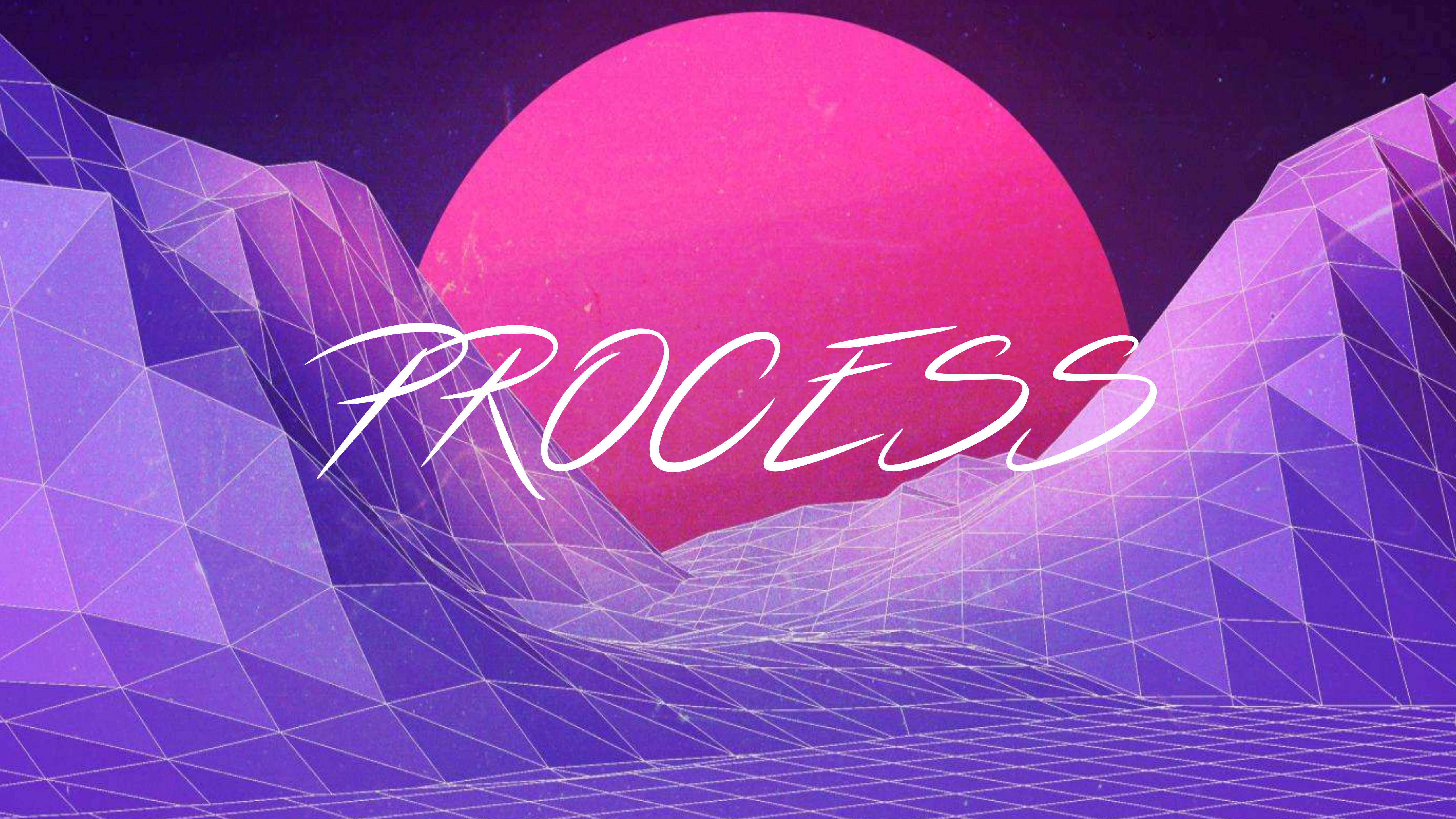
pipeline

controller

view

template

*RHODENIX
REQUEST*



PROCESS

LIGHTWEIGHT

System statistics / limit

Atoms:	22458 / 1048576 (2 % used)
Processes:	248 / 262144 (0 % used)
Ports:	8 / 65536 (0 % used)
ETS:	58 / 8192 (1 % used)
Distribution buffer busy limit:	1048576

ISOLATED

In simple relativity theory the concept of simultaneity just does not exist. The point is that in reality objects do not share state, I believe its not a good idea to model what cannot exist in reality in software ...

Everything that can be achieved with sharing and locks can also be achieved with pure message passing and no locks. The [sic] is the Erlang way.

EXECUTES

CODE

FOUR BLOCKS OF MEMORY

MAILBOX

HEAP

PCB

STACK

MAILBOX

Pid	Name or Initial Func	Reds	Memory	MsgQ	Current Function
<0.256.0>	Elixir.Plug.Supervisor	1	7104	0	gen_server:loop/7
<0.241.0>	ranch_sup	1	7104	0	gen_server:loop/7
<0.221.0>	hackney_sup	1	7104	0	gen_server:loop/7
<0.203.0>	Elixir.Logger.Supervisor	1	7104	0	gen_server:loop/7
<0.380.0>	erlang:apply/2	2581	8756	0	timer:sleep/1
<0.145.0>	Elixir.Hex.State	1	8844	0	gen_server:loop/7
<0.202.0>	application_master:start_it/4	1	8888	0	application_master:loop_it/4
<0.394.0>	observer_port_wx:init/1	2	8904	0	wx_object:loop/6
<0.247.0>	cowboy_clock	265	8932	0	gen_server:loop/7

PROCESS
CONTROL BOARD


```
    Uint32 rcount; /* suspend count */
    int schedule_count; /* Times left to reschedule a low prio
process */
    Uint reds; /* No of reductions for this process */
    Eterm group_leader; /* Pid in charge (can be boxed) */
    Uint flags; /* Trap exit, etc (no trace flags anymore) */
    Eterm fvalue; /* Exit & Throw value (failure reason) */
    Uint freason; /* Reason for detected failure */
    Eterm ftrace; /* Latest exception stack trace dump */
```

[HTTPS://GITHUB.COM/ERLANG/OTP/BLOB/OTP-21.0/ERTS/EMULATOR/BEAM/
ERL_PROCESS.H#L927](https://github.com/erlang/otp/blob/OTP-21.0/erts/emulator/beam/ERL_PROCESS.H#L927)

PROCESS STATES

FREE

WAITING

RUNNABLE

RUNNING

EXITING

GARBING

SUSPENDED

garbing computer science



All

Images

Videos

News

Shopping

More

Settings

Tools

About 120,000 results (0.45 seconds)

Did you mean: **garbage** computer science

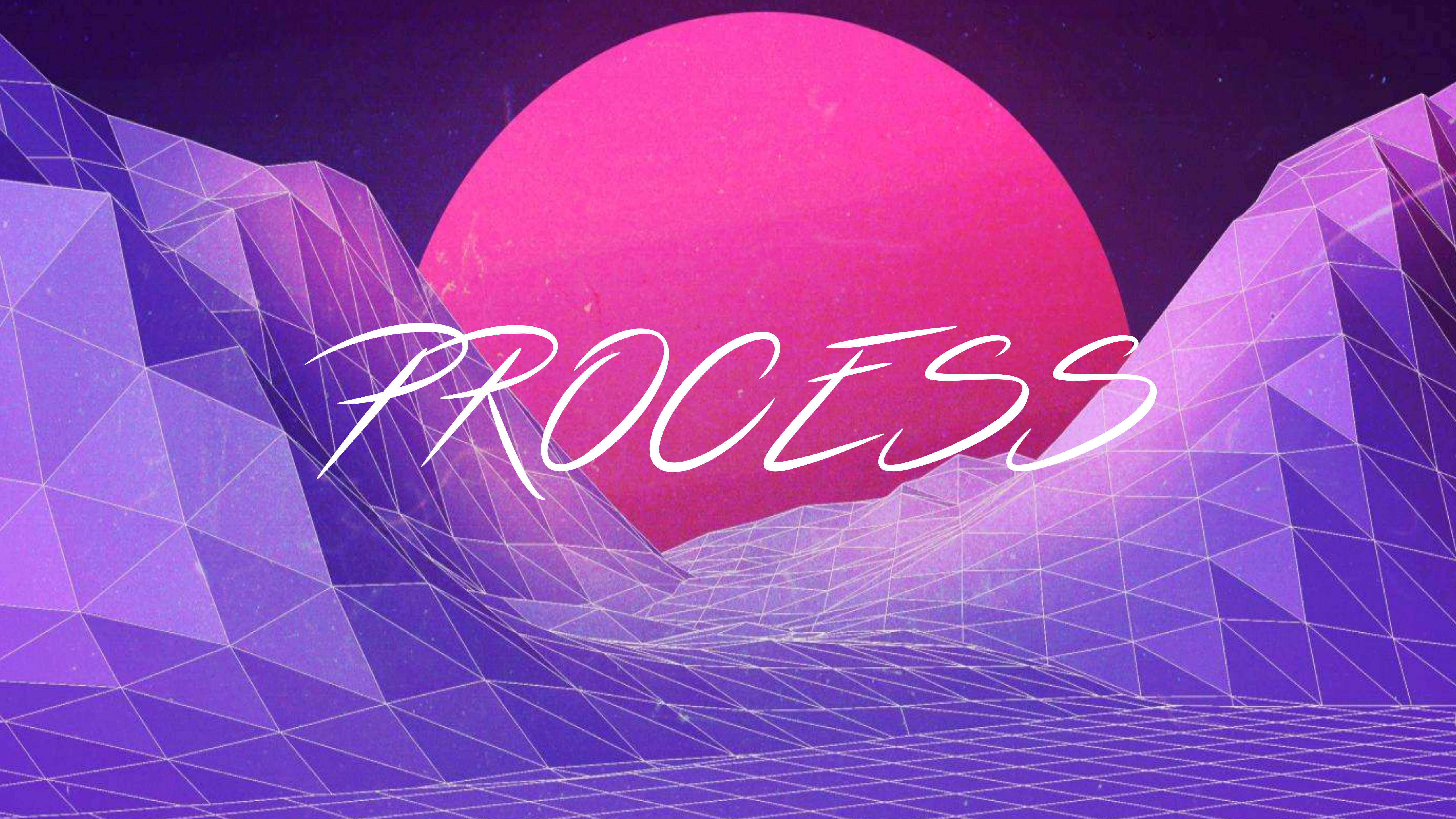
"NORMAL" STATES

WAITING

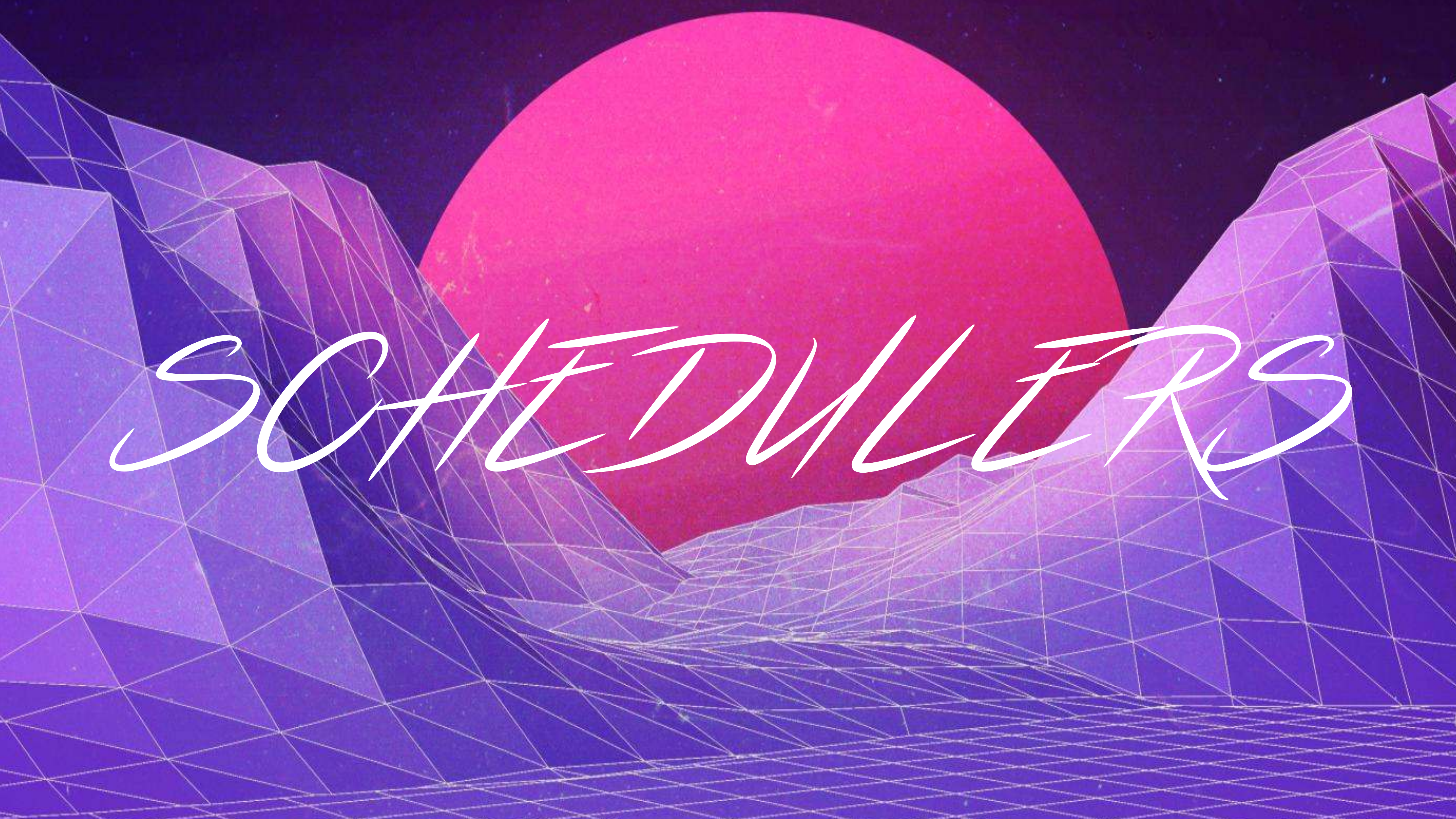
RUNNABLE

RUNNING


```
iex(5)> :erlang.process_info self(), :status  
{:status, :running}
```

PROCESS



SCHEDULERS



SINGLE SCHEDULER


```
iex(1)> :erlang.system_info:(smp_support)
```

```
true
```

```
iex(2)> :erlang.system_info :schedulers_online
```

```
4
```



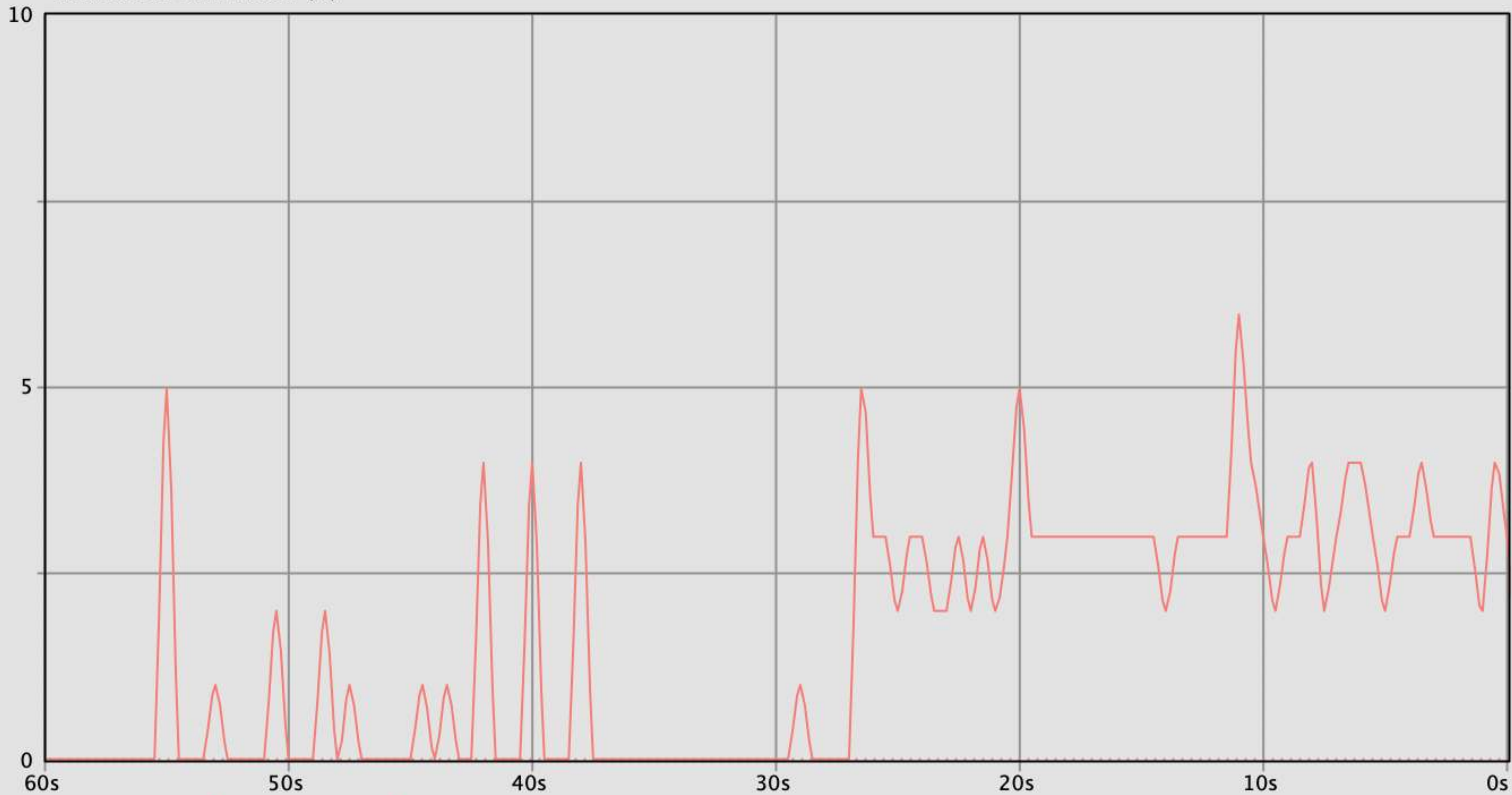
```
~> iex --erl "+S 1"
```

```
Erlang/OTP 21 [erts-10.0] [source] [64-bit] [smp:1:1] [ds:1:1:10] [async-threads:1] [hipe]
```

```
Interactive Elixir (1.7.3) - press Ctrl+C to exit (type h() ENTER for help)
```

```
iex(1)> :erlang.system_info :schedulers_online  
1
```


Scheduler Utilization (%)



Scheduler: 1 Dirty cpu: 1 (dotted)


```
iex(25)> Process.sleep(10_000)
[info] GET /
[debug] Processing with SimpleWeb.PageController.index/2
  Parameters: %{}
  Pipelines: [:browser]
[info] Sent 200 in 150μs
```



PREEMPTIVE*



REDUCTIONS



4000

REDUCTIONS

```
#DEFINE CONTEXT_REDS 4000 /* SWAP PROCESS OUT AFTER THIS NUMBER */
```



[HTTPS://GITHUB.COM/ERLANG/OTP/BLOB/MASTER/ERTS/EMULATOR/BEAM/ERL_VM.H#L39](https://github.com/erlang/otp/blob/master/erts/emulator/beam/erl_vm.h#L39)

RUN QUEUE

WAITING



RUNNABLE



RUNNING



SCHEDULER



RUN QUEUE

TASK 1

TASK 2

TASK N




```
iex(11)> :erlang.statistics(:run_queue)  
0
```


SCHEDULER



RUN QUEUE

TASK 1

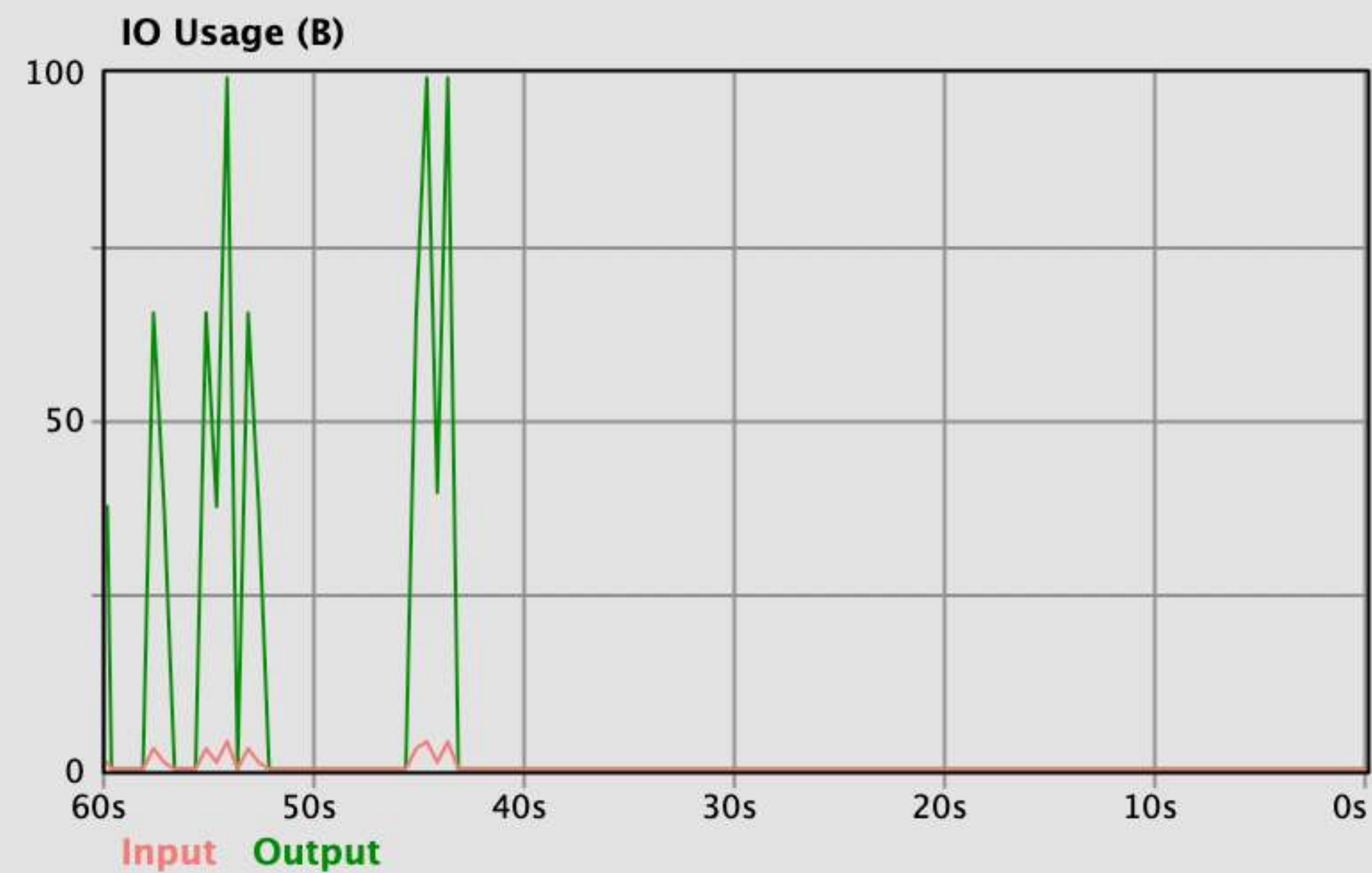
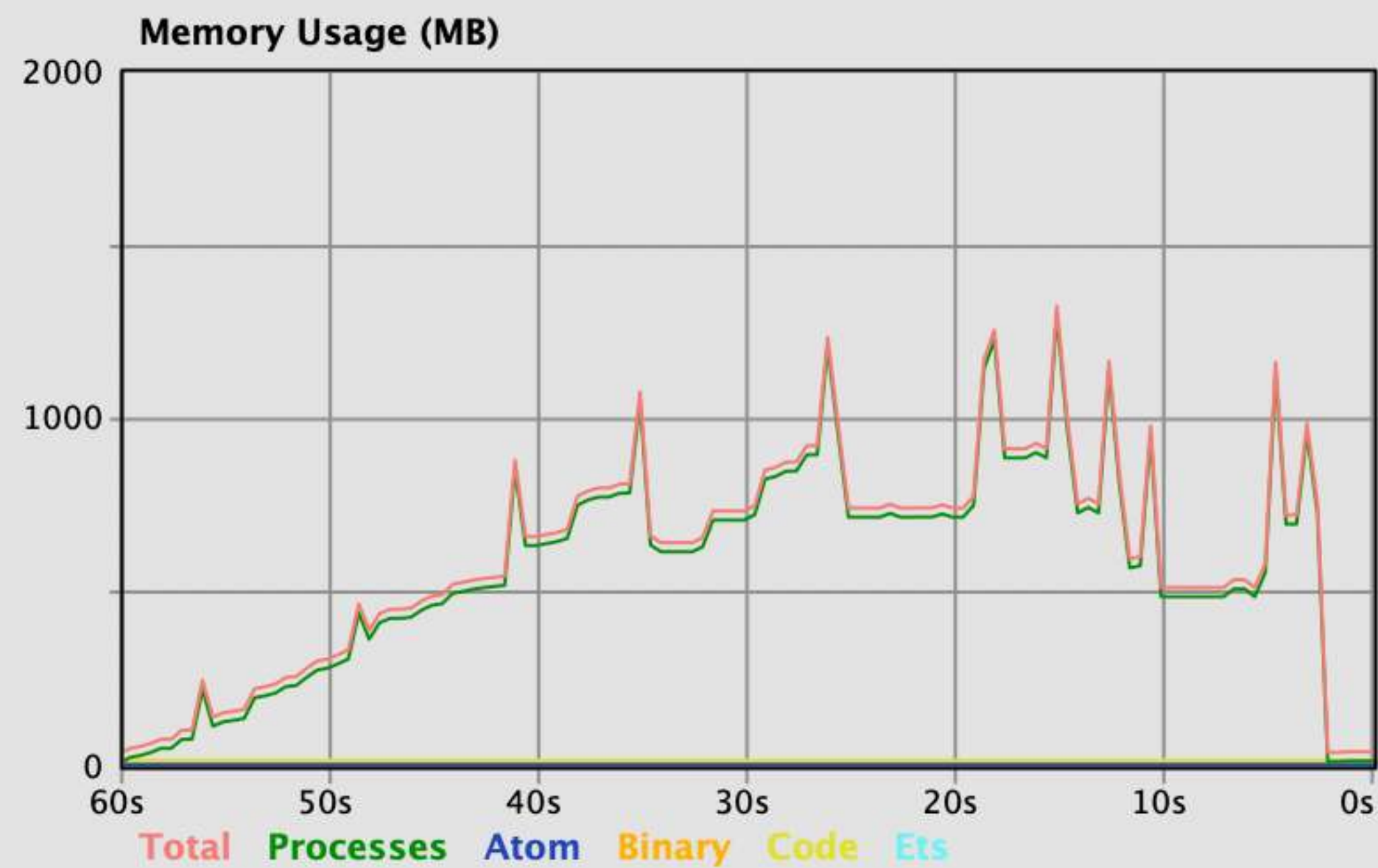
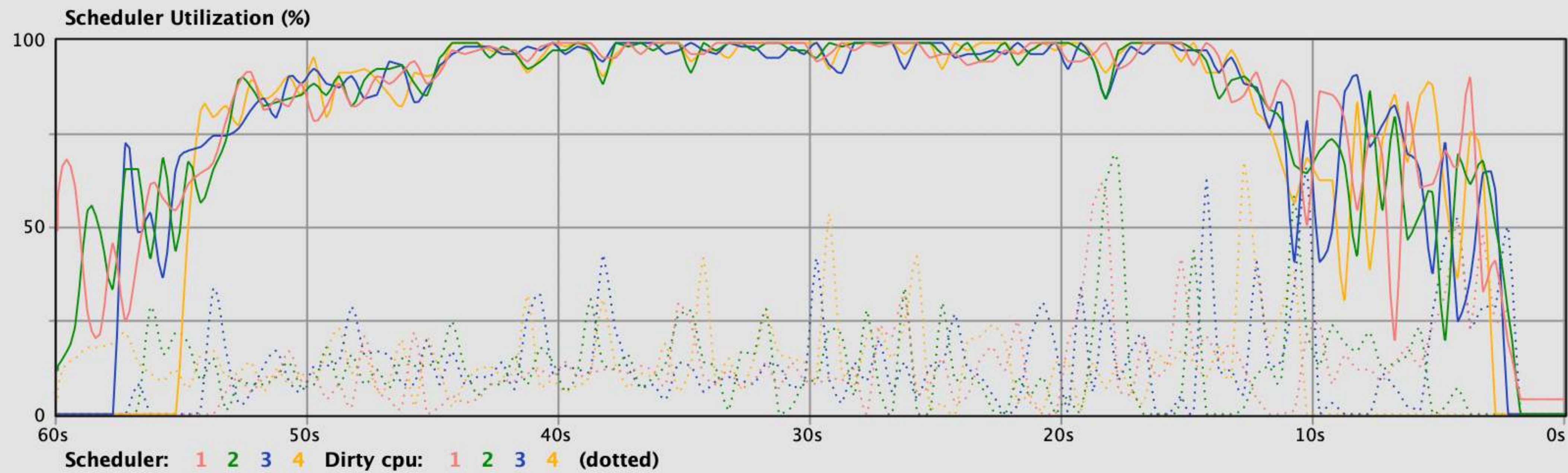
TASK 2

TASK N



The background features a dark purple field with a large, solid pink circle in the upper center. On either side of the circle are complex, low-poly geometric structures in shades of purple and blue, resembling crystalline or architectural forms. The foreground is filled with a dense, overlapping pattern of thin, light purple lines that create a sense of depth and movement.

MULTIPLE
SCHEDULES
(SNP)



RUN QUEUE

SCHEDULER



MAX PRIORITY

HIGH PRIORITY

NORMAL & LOW



LOAD BALANCING



ASK

STEALING



MIGRATION LOGIC



SCHEDULER 1



SCHEDULER 2



MIGRATION LOGIC

SCHEDULER 1

RUN QUEUE 1



SCHEDULER 2

RUN QUEUE 2



SCHEDULER N

RUN QUEUE N



FINELY

TWED



SCHEDULER FLAGS

`elixir --erl "+sbt ns" #etc`

[HTTP://ERLANG.ORG/DOC/MAN/ERL.HTML#FLAGS](http://erlang.org/doc/man/erl.html#flags)

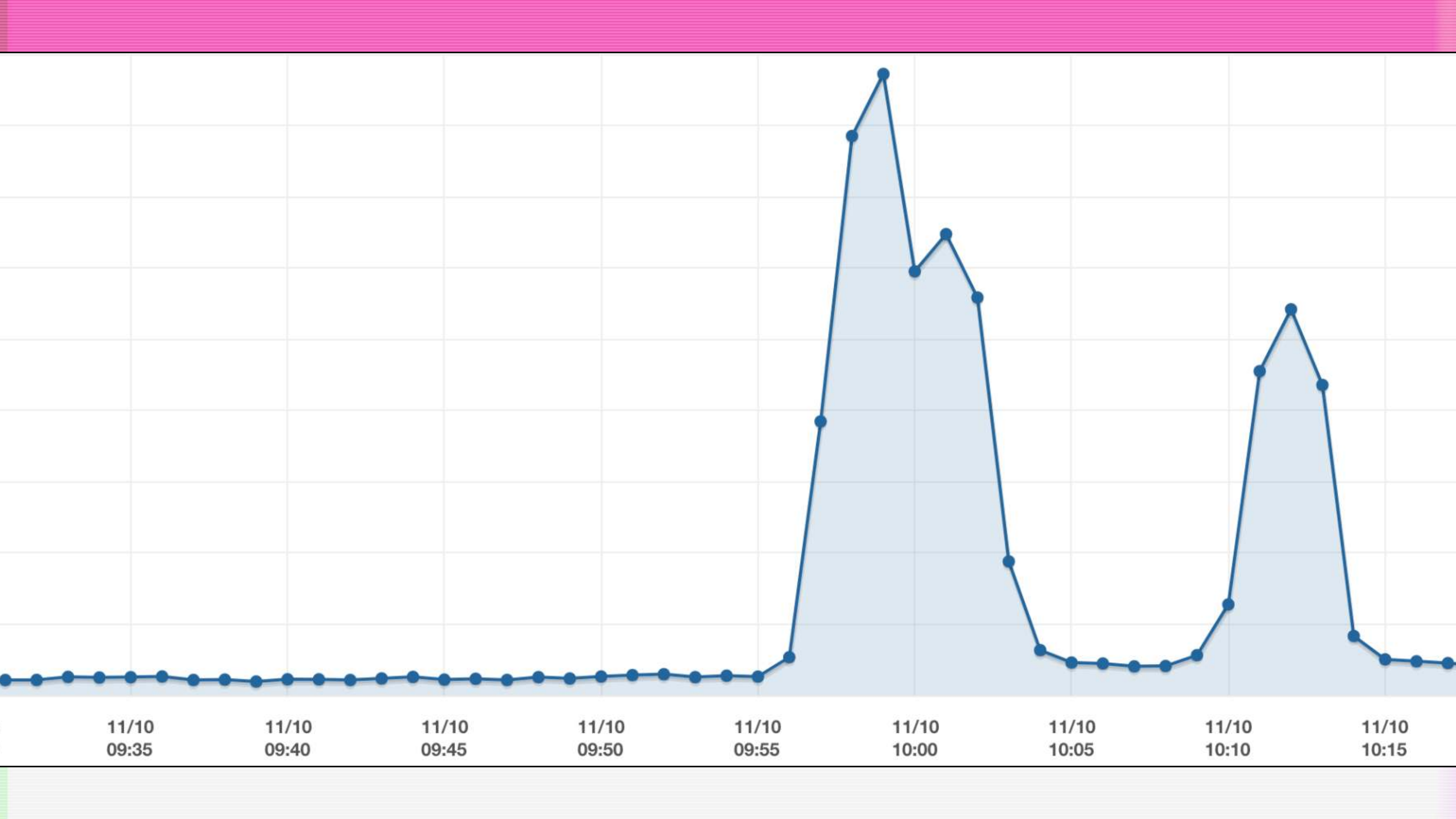
MOCHIT SCHEDULERS



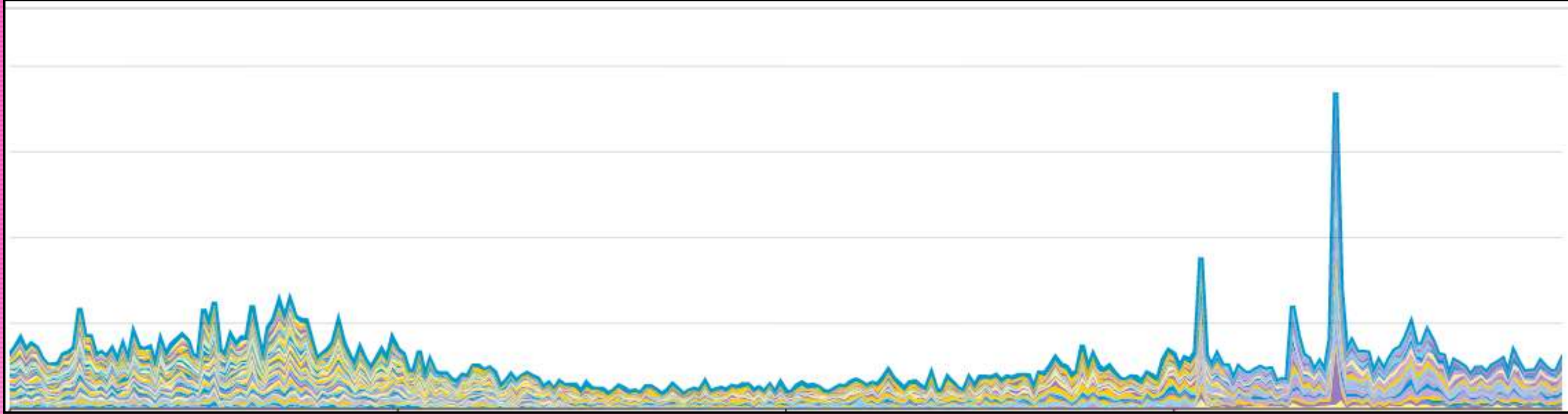


HOW HAS

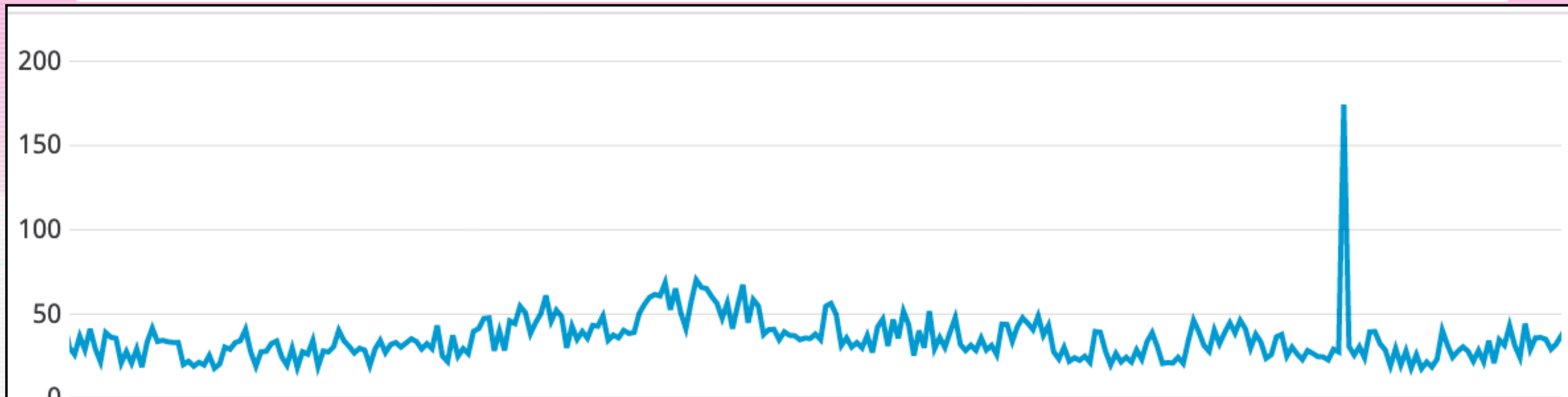
THIS FELT?



RUN QUEUE



LATENCY



REQUEST

SCHEDULER ASSIGNED

ADDED TO RUN QUEUE

TRANSFORMATIONS

RESPONSE

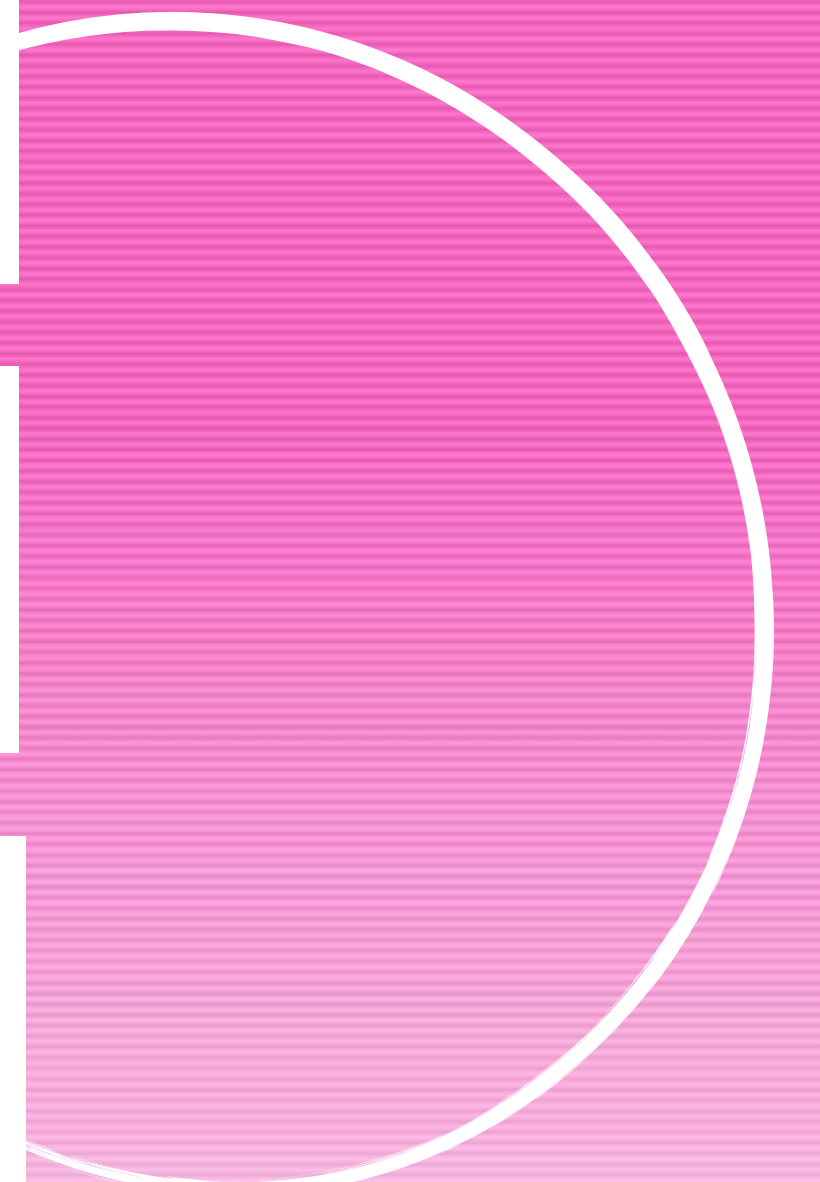


IMAGE CREDITS

<https://wallpaper-gallery.net/single/vaporwave-background-hd-14.html>

<http://www.desktopimages.org/preview/770453/1920/1080/>

[https://www.wallpapervortex.com/
wallpaper-52083_1_miscellaneous_digital_art_vaporwave_image_glitch.html](https://www.wallpapervortex.com/wallpaper-52083_1_miscellaneous_digital_art_vaporwave_image_glitch.html)

<https://www.deviantart.com/epix0r/art/R-o-a-d-W-a-y-1920-x-1080-HD-629299503>

<https://wallpaper-gallery.net/single/glitch-wallpaper-hd-11.html>

RESOURCES

The BEAM Book - <https://github.com/happi/theBeamBook>

<https://jlouisramblings.blogspot.com/2013/01/how-erlang-does-scheduling.html>

<http://www.erlang-factory.com/upload/presentations/105/KennethLundin-ErlangFactory2009London-AboutErlangOTPandMulti-coreperformanceinparticular.pdf>

<https://hamidreza-s.github.io/erlang/scheduling/real-time/preemptive/migration/2016/02/09/erlang-scheduler-details.html>

DZIEKUJE

@bgmarx